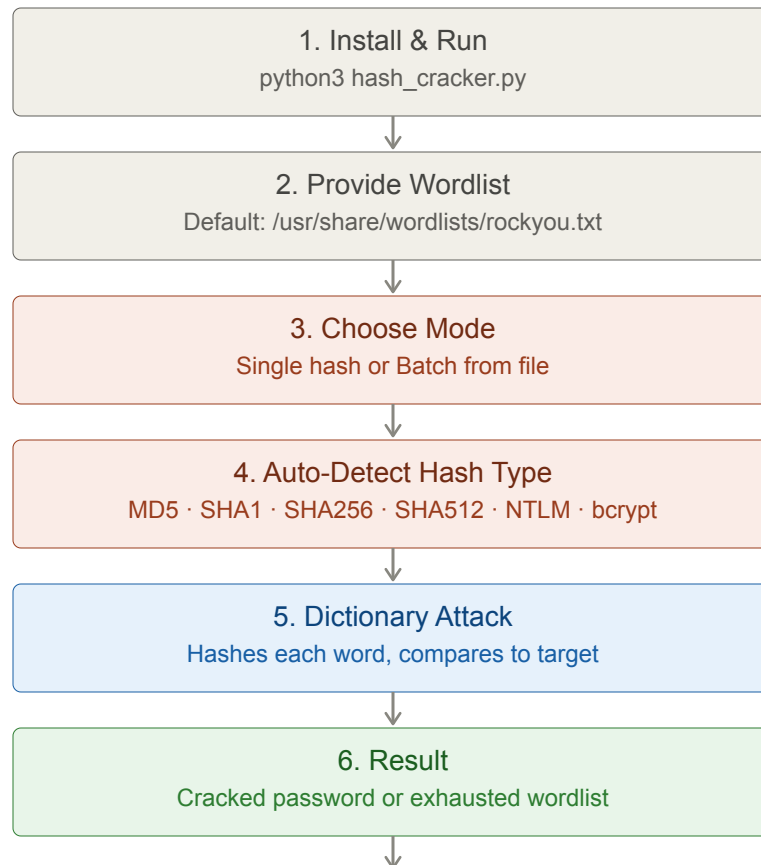


Hash Cracker — cheat sheet

Dictionary attack tool for cracking password hashes. Auto-detects MD5, SHA1, SHA256, SHA512, NTLM, and bcrypt.



Step 1 — Install & Run

```
pip install -r requirements.txt
python3 hash_cracker.py
```

Step 2 — Choose a Wordlist

```
# Built-in on Kali / Parrot
/usr/share/wordlists/rockyou.txt      (14M passwords)
/usr/share/wordlists/fasttrack.txt    (222 common passwords)

# Download larger lists
https://github.com/danielmiessler/SecLists (many options)
```

Press Enter at the wordlist prompt to use the rockyou.txt default.

Step 3 — Two Modes

Mode	When to use
1 – Single hash	You have one hash to crack — paste it manually
2 – Batch mode	You have a file of hashes (one per line) — e.g. from /etc/shadow dump

Step 4 — Supported Hash Types (Auto-Detected)

Type	Example hash (truncated)	Common source
MD5	5f4dcc3b5aa765d61d...	Old web apps, databases
SHA1	5baa61e4c9b93f3f...	Git, older systems
SHA256	5e884898da2804...	Modern Linux shadow files
SHA512	b109f3bbbc244...	Modern Linux shadow files (preferred)
NTLM	8846f7eaae8fb1...	Windows SAM / Active Directory
bcrypt	\$2b\$12\$...	Modern web apps (slow to crack)

Step 5 — Reading Results

```
[*] Detected hash type: MD5
[*] Cracking: 5f4dcc3b5aa765d61d8327deb882cf99
[*] Speed: 45,231 attempts/sec

[+] CRACKED in 0.07s after 3,241 attempts!

Hash      : 5f4dcc3b5aa765d61d8327deb882cf99
Password  : password
```

Batch mode prints a summary table with cracked / failed counts after processing all hashes.

Key idea: bcrypt is intentionally slow — cracking even a weak bcrypt password can take hours. For NTLM hashes from Windows, rockyou.txt is usually enough for weak passwords. For SHA256/SHA512 from /etc/shadow, add rules with hashcat for better coverage.